Practice workbook · Explain homeostasis and negative feedback

Defining homeostasis, explaining negative feedback with temperature or glucose examples, and why a constant internal environment matters for enzymes.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-HOMEOSTASIS-NEGATIVE-FEEDBACK

1. Question 4 · part c · 4 marks

January 2022 Paper 1

Worked-example retry · Attempt part c only.

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4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

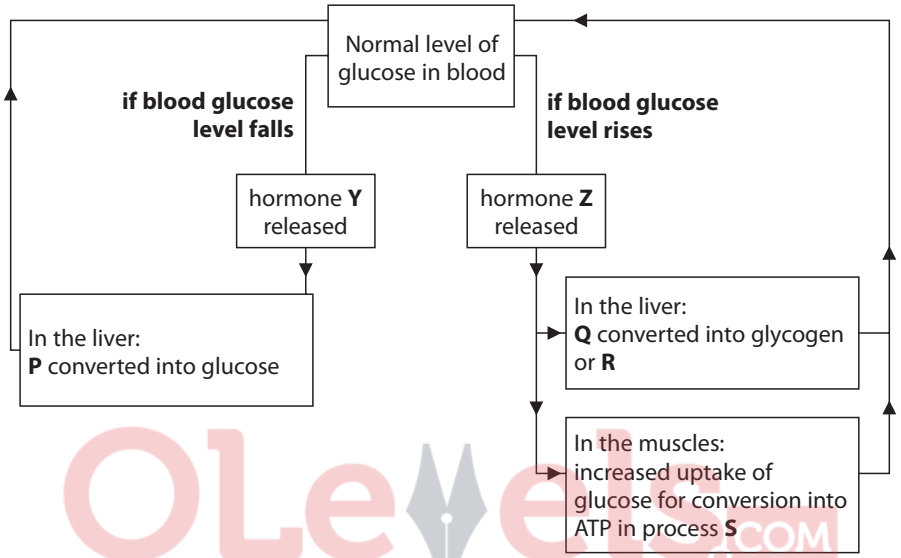
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1. Question 4 · part c · 4 marks

January 2022 Paper 1

Worked-example retry · Attempt part c only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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1. Question 4 · part c · 4 marks

January 2022 Paper 1

Worked-example retry · Attempt part c only.

(iii) Identify P, Q, R and S.

(4)

P .....

Q .....

R .....

S .....

(c) The homeostatic process shown in the diagram is controlled by negative feedback.

Using the example shown in the diagram, explain what is meant by negative feedback.

(4)

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(Total for Question 4 = 13 marks)

1. Question 4 · part c · 4 marks · Working and answer

Working and answer

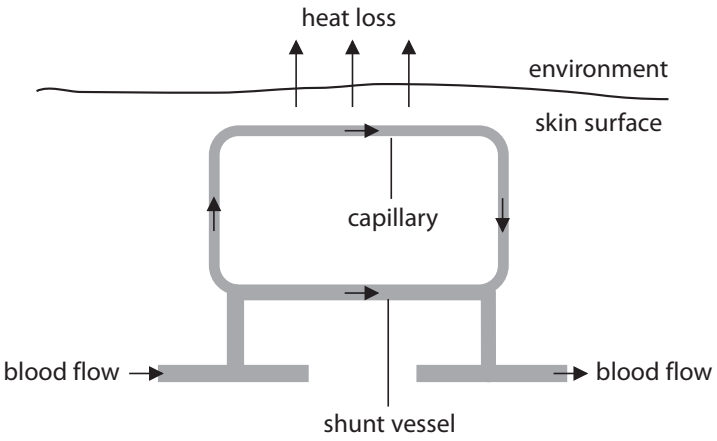
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2. Question 7 · part c · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

7 The diagram shows a section through skin on a cold day.



(a) Draw a diagram in the space to show how the section of skin shown in the diagram would be different on a hot day.

(2)

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2. Question 7 · part c · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

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(b) Explain why the section of skin shown in the diagram would be different on a hot day.

(4)

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(c) Explain how the control of body temperature is an example of negative feedback.

(2)

(Total for Question 7 = 8 marks)

2. Question 7 · part c · 2 marks · Working and answer

Working and answer

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3. Question 4 · part a · 3 marks

January 2022 Paper 1

Further practice · Attempt part a only.

4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

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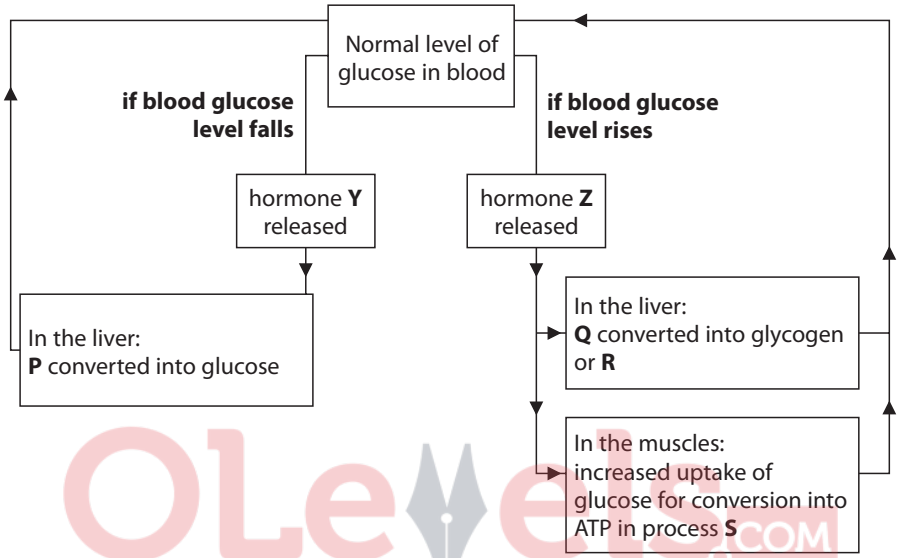
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3. Question 4 · part a · 3 marks

January 2022 Paper 1

Further practice · Attempt part a only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

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P 6 7 0 5 9 A 0 1 2 2 4

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

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4. Question 3 · part d · 3 marks

January 2022 Paper 2

Further practice · Attempt part d only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

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2 .....

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(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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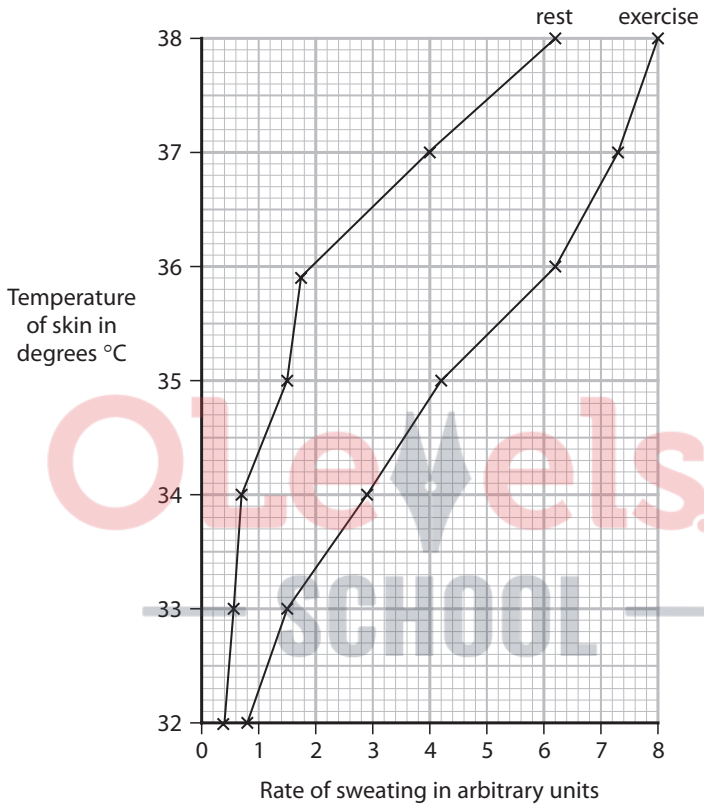
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4. Question 3 · part d · 3 marks

January 2022 Paper 2

Further practice · Attempt part d only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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4. Question 3 · part d · 3 marks

January 2022 Paper 2

Further practice · Attempt part d only.

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(i) Describe how the skin temperature varies as the rate of sweating changes for the person at rest.

(2)

(ii) Suggest a method that could be used to measure the rate of sweating.

(2)

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(d) Explain the importance of maintaining a constant body temperature.

(3)

(Total for Question 3 = 13 marks)

4. Question 3 · part d · 3 marks · Working and answer

Working and answer

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5. Question 8 · part b.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

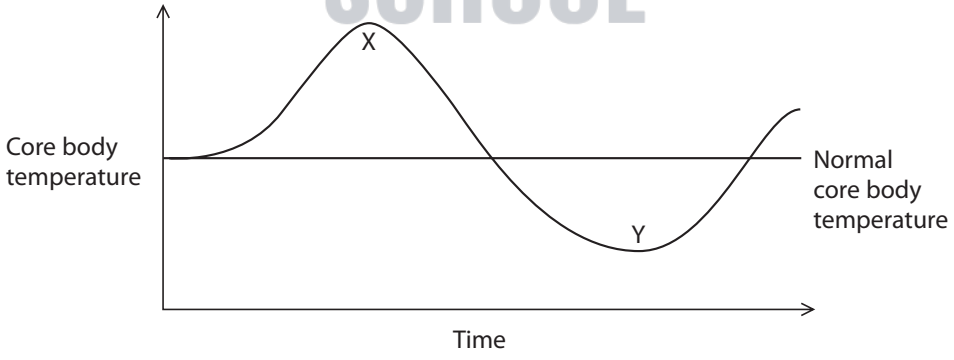
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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P 7 4 4 9 5 A 0 2 4 2 8

5. Question 8 · part b.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

(iii) Explain why the control of body temperature is an example of negative feedback.

(2)

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5. Question 8 · part b.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

(c) Hypothermia is a condition where the core body temperature drops below 35 °C.

Explain why it is important to increase the core body temperature of people with hypothermia to within the normal range.

(5)

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(Total for Question 8 = 14 marks)

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5. Question 8 · part b.iii · 2 marks · Working and answer

Working and answer

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6. Question 3 · part a · 4 marks

June 2025 Paper 1

Further practice · Attempt part a only.

3 The control of body temperature is achieved by a process known as homeostasis.

(a) Explain what is meant by homeostasis.

(4)

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6. Question 3 · part a · 4 marks

June 2025 Paper 1

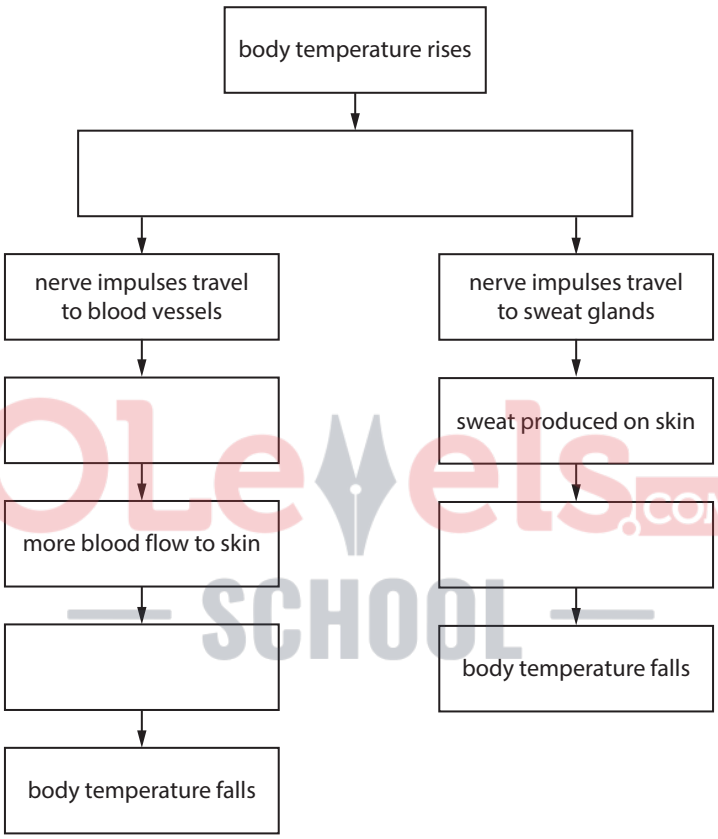
Further practice · Attempt part a only.

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(b) The diagram shows some of the reactions of the body to an increase in temperature.



Complete the diagram by putting in a suitable word or sentence in each of the blank boxes.

(6)

6. Question 3 · part a · 4 marks · Working and answer

Working and answer

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